

Second Terminal Examination - 2079

Class : XII

Subject: Physics (1021)

Full Marks: 75

Time: 3 hrs.

SET 'B'

Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

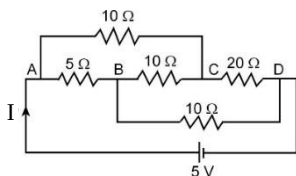
Attempt all the questions.

Group A

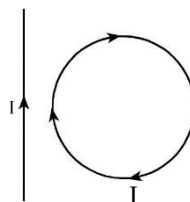
Rewrite the best alternative of each question in your answer sheet. [11 × 1 = 11]

- A mass m is vertically suspended from spring of negligible mass and system oscillates with frequency f . If the mass $4m$ is suspended from the same spring, the frequency will be
 a) $\frac{f}{4}$ b) $\frac{f}{2}$ c) $2f$ d) $4f$
- The maximum acceleration of a body executing SHM is a_0 and the maximum velocity is v_0 . The amplitude of motion is given by
 a) $\frac{v_0^2}{a_0}$ b) $\frac{v_0}{a_0}$ c) $\frac{a_0^2}{v_0}$ d) $\frac{1}{v_0 a_0}$
- Radius of gyration of a uniform rod of length 'L' about perpendicular axis through its middle is
 a) $\frac{L}{\sqrt{3}}$ b) $\frac{L}{\sqrt{12}}$ c) $\frac{L}{\sqrt{2}}$ d) $\frac{L}{\sqrt{8}}$
- Which relation is correct for isothermal process?
 a) $\Delta Q = \Delta U$ b) $\Delta W = \Delta U$
 c) $\Delta Q = \Delta W$ d) none of these
- A Carnot engine operates at 800 K and 200 K. If the engine takes 8 J of heat, then what is the amount of work done by the engine?
 a) 6 J b) 4 J c) 10 J d) 2 J

6. What type of vibrations are produced in a sitar wire?
 a) progressive, transverse b) progressive, longitudinal
 c) transverse, stationary d) longitudinal, stationary
7. With the increase in temperature, the frequency of sound from an open organ pipe
 a) increases b) decreases
 c) remains constant d) may increase or decrease
8. The current 'I' drawn from the 5 V source will be



- a) 0.67 A b) 0.17 A c) 0.33 A d) 0.5 A
9. A moving coil galvanometer has a resistance of 90Ω . If only 10% of the main current may flow through the galvanometer, in what way and which resistance is to be used?
 a) 10Ω in series b) 810Ω series
 c) 10Ω in parallel d) 810Ω in parallel
10. In the given figure, the loop is fixed but straight wire can move. The straight wire will
 a) remain stationary
 b) move towards the loop
 c) move away from the loop
 d) rotate about the axis.



11. e/m of electron was determined by
 a) William Crooke b) Goldstein
 c) J.J. Thomson d) Millikan

Group - B

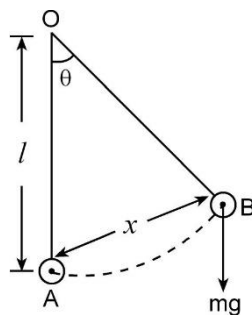
Give short answer to the following questions.

[8 × 5 = 40]

12. a) State the principle of conservation of angular momentum. [1]
- b) If the earth were to shrink suddenly, what will happen to the length of day? Explain. [1]
- c) A constant torque of 200 Nm turns a wheel about its center. The moment of inertia of it about the axis is 100 kgm^2 . Find the kinetic energy gained after 20 revolutions starting from rest. [3]

OR

- a) Show that angular momentum of a rigid body about rotating axis is $L = I\omega$, where the symbols carry their usual meanings. [3]
- b) A thin uniform rod is rotated first about an axis through its center and perpendicular to its length with certain angular velocity. The same rod is again rotated with same angular velocity about an axis passing through its one end and perpendicular to its length. Find the ratio of angular momenta of rods in these two cases. [2]
13. Figure shows a bob of mass ' m ' suspended by a weightless string of effective length ' l '. When the bob is displaced through a small displacement of ' x ' and released, it will oscillate freely.



- a) Complete the figure and show that motion of bob is simple harmonic. [2]
- b) Obtain an expression for the period of oscillation of the bob. [1]
- c) What will happen if the angular displacement ' θ ' of bob is made large? [1]

- d) If we perform the experiment of simple pendulum inside water, what will happen to the motion? Outline the two major points. [1]

14. a) In an adiabatic process, the assumption of no heat transfer is very important, since we can use the adiabatic approximation only in very rapid processes. Deduce the adiabatic equation $PV^\gamma = \text{constant}$, where the symbols have their usual meaning. [3]

- b) Air is compressed adiabatically to half of its volume. Calculate the percentage change in temperature. (Given, $\gamma = 1.4$) [2]

OR

- a) How is the Petrol engine different from a Diesel engine? [2]

- b) A refrigerator transfers heat from a cold body to a hot body. Does this not violate the second law of thermodynamics? [1]

- c) Assuming the domestic refrigerator as a reversible engine working between the melting point of ice and the room temperature of 27°C , calculate the energy in Joule that must be supplied to freeze one kg of water. (Given, the temperature of water is 0°C , $L = 80 \text{ cal g}^{-1}$). [2]

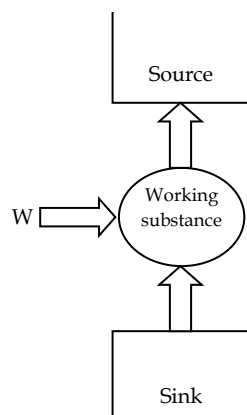


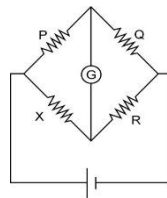
Fig. Refrigerator

15. a) Discuss the effect of temperature and pressure for the velocity of sound in gas and also show graphical sketch. [2]

- b) Calculate the frequency of a monotonous sound produced by 30 cm long flute open at both ends and being played in fundamental mode (first harmonics). The air column inside flute has temperature 27°C and the velocity of sound in air at 0°C was 332m/s. [2]

- c) The velocity of sound increases on cloudy day. Why? [1]

16. a) In 1845, a German physicist, Gustav Kirchhoff, developed a pair of laws that deal with the conservation of charge and energy within electrical circuits. The figure shows a Wheatstone bridge circuit.



Prove that $\frac{P}{Q} = \frac{X}{R}$ at null deflection of galvanometer.

[3]

- b) Equal currents are flowing through two infinitely long parallel wires. What will be the magnetic field at a point mid-way, when the currents are flowing (i) in the same direction (ii) in the opposite direction? [2]
17. a) Derive the formula for the magnetic field at the centre of a circular coil carrying current. Explain why the magnetic field at the centre of the coil disappears when the circular coil is made infinitely large. [3]
- b) The plane of a $5 \text{ cm} \times 8 \text{ cm}$ rectangular loop of wire is parallel to a 0.19 T magnetic field. The loop carries a current of 6.2 A . (a) What is the torque acting on the coil? (b) What is the maximum torque on it? [2]
18. a) Beams of electrons and protons having the same initial K.E. enter normally into an electric field, which beam will more curved? Justify. [2]
- b) An electron having 450 eV of energy moves at right angles to a uniform magnetic field of flux density $1.50 \times 10^{-3} \text{ T}$. Find the radius of its circular orbit. (Given, specific charge of the electron is $1.76 \times 10^{11} \text{ C/kg}$) [3]
19. a) What is threshold frequency for photoelectric emission? Does it depends on the intensity of light? [2]
- b) Electrons with maximum kinetic energy of 3 eV are ejected from a metal surface by ultraviolet radiation of wavelength $1.5 \times 10^{-7} \text{ m}$. Determine work function, threshold wavelength and the stopping potential for the metal (Planck's constant, $h = 6.62 \times 10^{-34} \text{ Js}$) [3]

OR

- a) The total energy of an electron in an atom in any orbit is $E_n = -\frac{13.6\text{eV}}{n^2}$.

What does this negative energy indicate? [1]

- b) Now, the electron in a given Bohr's orbit has a total energy of -1.5 eV . Calculate its

(i) kinetic energy. [1]

(ii) potential energy. [1]

(iii) the wavelength of radiation emitted, when this electron makes a transition to the ground state. [2]

(Given, energy in the ground state = -13.6 eV and Rydberg's constant = $1.09 \times 10^7\text{ m}^{-1}$)

Group - C

Give long answer to the following questions.

[3 × 8 = 24]

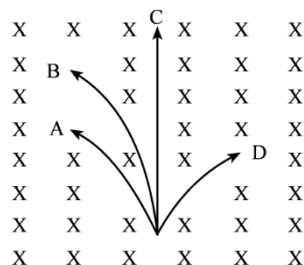
20. a) What is standing wave? Using the principle of superposition, find the position of nodes and antinodes. Also, show that distance between two consecutive nodes is equal to the distance between two consecutive antinodes. [4]
- b) Sketch the wave pattern for the formation of stationary wave in an open organ pipe played in first overtone. [1]
- c) A train is approaching a cliff at 20 m/s . The driver sounds a whistle of frequency 800 Hz . What will be the frequency of echo as heard by the driver? The velocity of sound in air is 350 m/s . [2]
- d) Sound is a pressure wave. Comment on this. [1]

OR

- a) A plane progressive wave is represented by $y = 0.1 \sin\left(200\pi t - \frac{20\pi x}{17}\right)$, where y is in mm, t is in seconds and x in meters. Calculate:
- i) Phase velocity of wave and maximum particle velocity. [2]
- ii) The phase difference between two points 0.25 m apart in medium. [1]

- b) What is the fundamental difference between simple harmonic motion and wave-motion? [1]
- c) Derive an expression for the apparent frequency received by an observer when it moves towards the stationary source. [3]
- d) A sound source and observer both are at rest on the earth but a strong wind is blowing. Is there a Doppler effect? Explain. [1]

21. a) A neutron, a proton, an electron and an α -particle enter a region of constant magnetic field with equal velocities. The magnetic field is along the inward normal to the plane of the paper. The tracks of the particles are labelled in Fig. Relate the tracks to the four particles. [3]



- b) Hall Effect Principle is one of the most popular theories in magnetic field measurement. Only two decades later, with the introduction of semi-conductors, the work and uses of Hall Effect was utilized in an effective manner than for good conductors for comparable currents, fields and dimensions. Why? [2]
- c) A slice of indium antimonide is 2.5mm thick and carries a current of 150 mA. A magnetic field of flux density 0.5 T, correctly applied, produces a maximum Hall voltage of 8.75 mV between the edges of the slice. Calculate the number of free charge carriers per unit volume, assuming they each have a charge of $1.6 \times 10^{-19} \text{C}$. [3]

OR

- a) What is thermoelectric e.m.f.? [1]
- b) What are the two factors on which thermoelectric e.m.f. produced in a thermocouple depends? [1]

c) The potentiometer is a device that was developed in the early days of electronics development. How can you use to determine the internal resistance of a cell? [3]

d) A copper wire has 1.0×10^{29} free electrons per cubic metre, a cross sectional area of 2.0 mm^2 and carries a current of 5.0 A. Calculate the force acting on each electron if the wire is now placed in a magnetic field of flux density 0.15 T which is perpendicular to the wire.

($e = -1.6 \times 10^{-19} \text{ C}$) [3]

22. a) What are Bohr's postulates of hydrogen atom? Derive an expression for the radius of n^{th} of orbit hydrogen atom. [1+3]

b) What do you meant by de-Broglie waves? [1]

c) Obtain the de Broglie wavelength of neutron of kinetic energy 150 eV.

(Given, mass of neutron = $1.675 \times 10^{-27} \text{ kg}$,

Planck's constant = $6.6 \times 10^{-34} \text{ Js}$, $1\text{eV} = 1.6 \times 10^{-19} \text{ C}$) [3]

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